Relational Schema
Supermarket Management System

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Relational Schema

The relational schema represents the database design for the supermarket management system, converted from the ER model into a set of relations (tables). It consists of eight main tables: Category, Customer, Employee, Supplier, Item, Bill, Bill_Details, and Procures.

Each table has a primary key that uniquely identifies its records, while foreign keys establish relationships between different tables. For example, `Category_ID` connects Employee and Item with the Category table, while `Customer_ID` and `Employee_ID` connect Bill with Customer and Employee.

The Bill_Details and Procures tables act as junction tables to represent many-to-many relationships, using composite primary keys formed from the participating entities.

Composite attributes such as customer name and address have been divided into atomic attributes, while the derived GST percentage is not stored because it can be obtained from the HSN code.

Overall, the schema maintains referential integrity and provides a normalized structure for efficiently managing customers, employees, inventory, suppliers, purchases, and billing information.

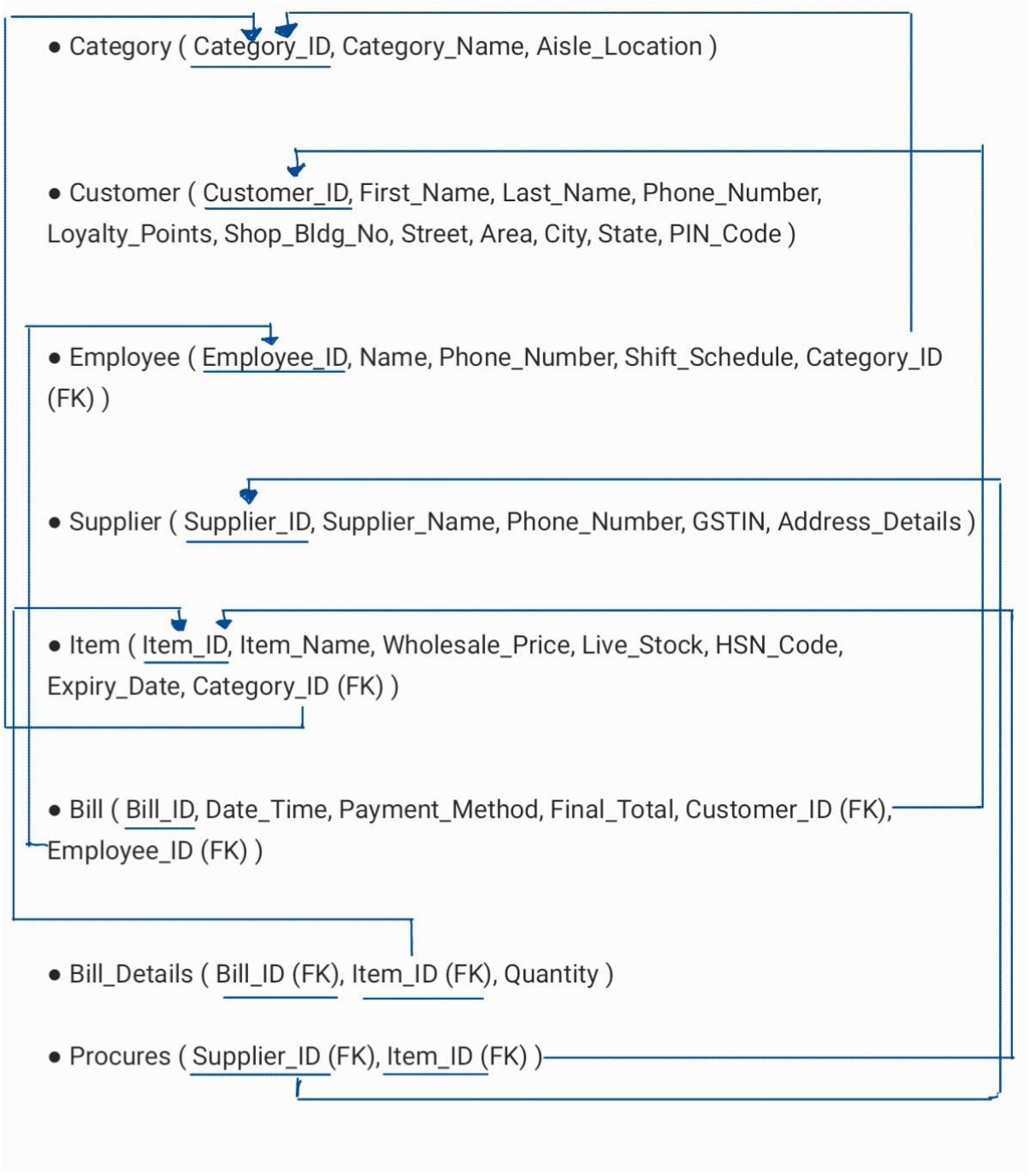


Figure 1: Supermarket Management System